

One gauge for every record IAM reads

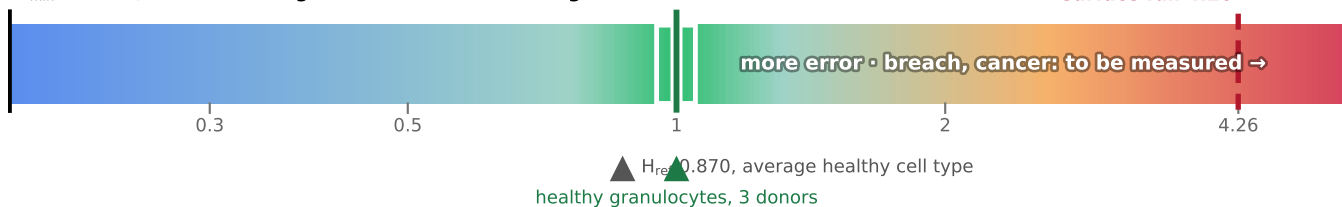
1 = healthy, in the middle · $H_{\min}$ = the floor where thermal kicks win; nothing reads below it · above 1 = more error

Cell · neutrophil · IAM-A

copy error at each CpG, read on single molecules

$H_{\min}$ 1×10^{-7} , off the left edge · (thermal kicks win against one ATP)

surface full 4.26

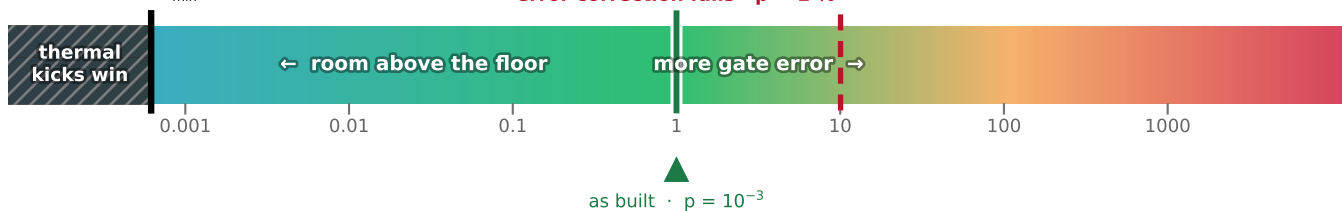


Qubit · transmon reading · two-qubit gate

$\varepsilon = -\ln(1 - p)$; 5 GHz at its own 35 mK, T_1 68 μ s, 40 ns gate, $p = 10^{-3}$

$H_{\min}$ 6.2×10^{-4} · thermal floor

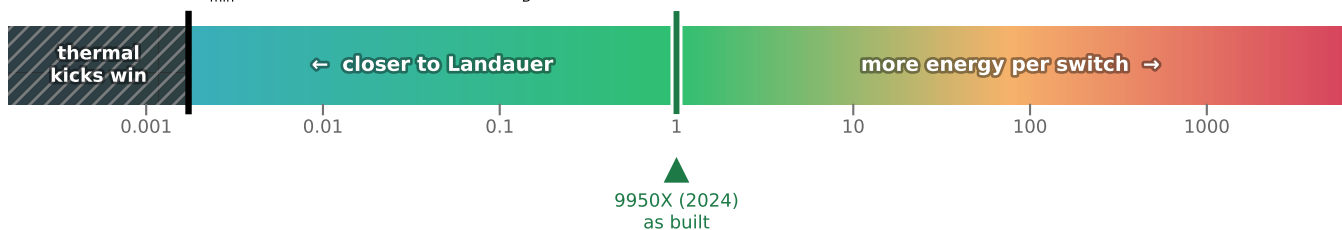
error correction fails · $p = 1\%$



Chip · AMD Ryzen 9 9950X · switching energy

$E = TDP / (N f)$; the chip as built against $k_B T_j \ln 2$

$H_{\min}$ 0.0017 · Landauer floor $k_B T \ln 2$



A = reading ÷ the same system's healthy reading (log scale, each gauge centred on 1)